Phase 2 Acceptance Demo - Audit Planning & Risk Assessment

End-to-end acceptance run for Phase 2 on the **deployed** app. Passing this is the gate before Phase 3 starts (per `checklist.md`: *"do not start a phase until the previous phase's acceptance criteria pass"*).

Acceptance criterion (checklist.md): *A demo engagement has an imported TB, locked materiality, documented risks linked to programs, and lead schedules with correct totals.*

Each step below has a **Do** (the action in the UI) and a **Validate** (what must be true for the step to pass). Tick the box only when Validate passes.

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0. Prerequisites

- [ ] App is reachable: `https://dn46rvtw2v7wc.cloudfront.net` loads the SPA and `/api/health` returns `200` with DB connectivity.
- [ ] **Two signed-in test users with roles assigned in Clerk** (this is the Phase 1 carry-over item - without it nothing below works):
- One **Partner** or **Admin** (does the approvals/locking).
- One **Auditor** or **Manager** (does the data entry; also used for the negative RBAC check in step 2).
- Assign via the **Users** screen (Admin only) or Clerk `public_metadata`.
- [ ] A **client** and an **engagement** exist and the engagement is **not completed** (create via Clients -> Engagements if needed). Note its URL - every Phase 2 screen hangs off `/engagements/<engagementId>/...`.
- [ ] Test data file: `docs/sample-trial-balance.csv` (balanced; current year sums to zero, prior year sums to zero).

> Sign convention: **debits positive, credits negative**; a valid TB sums to zero. The import rejects a TB that does not balance.

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1. Trial balance import

Screen: Engagement detail -> **Trial balance** (`/engagements/:id/trial-balance`). Open to all staff roles (Admin/Partner/Manager/Auditor); import/regroup needs Admin/Partner/Manager.

Do 1. Upload `docs/sample-trial-balance.csv`. The wizard calls `POST .../trial-balance/preview` and shows detected columns + a preview. 2. Map columns: **Account Code -> code**, **Account Name -> name**, **Current Year -> balance**, **Prior Year -> prior-year balance**. (Leave debit/credit unset - this file uses a single signed balance column.) 3. Confirm import (`POST .../trial-balance`). 4. Review auto-suggested FS areas; **refine grouping** for any line the keyword matcher left blank or mis-grouped (`PATCH .../trial-balance/fs-areas`).

Validate - [ ] Import **succeeds** (no "Trial balance does not balance" error) - proves the ±0 balancing rule passed. - [ ] 10 account lines imported. - [ ] FS-area summary (`GET .../trial-balance/summary`) shows the lines grouped, with these expected areas after refinement: - PPE -> `property_plant_equipment`, Inventory -> `inventory`, Trade receivables -> `trade_receivables`, Cash and bank -> `cash_and_bank`, Trade payables -> `trade_payables`, Borrowings -> `borrowings`, Share capital... -> `equity`, Revenue -> `revenue`, Cost of sales -> `cost_of_sales`, Operating expenses -> `operating_expenses`. - [ ] Prior-year column is populated (drives the comparison + lead-schedule movement). - [ ] **Negative check:** re-importing with one line edited so the TB no longer sums to zero returns a 422 validation error naming the difference. (Optional but recommended - proves errors propagate, not swallowed.)

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2. Materiality - save, then approve & lock

Screen: Engagement -> **Materiality** (`/engagements/:id/materiality`). Edit needs Admin/Partner/Manager; **approve/reopen needs Admin/Partner only**.

Do (as Auditor/Manager first, for the RBAC check) 1. As the **Auditor**, open Materiality and attempt to **Approve**. Expect it to be blocked (button hidden/disabled, or `POST .../materiality/approve` -> 403). 2. As **Manager or Partner**, fill the assessment (`PUT .../materiality`): - Benchmark: **Revenue**; benchmark amount: **900000** (= revenue from the TB). - Overall materiality %: e.g. **5** -> overall **45,000**. - Performance materiality %: e.g. **75** of overall -> **33,750**. - Clearly-trivial threshold %: e.g. **5** of overall -> **2,250**. - Save. 3. As **Partner/Admin**, **Approve** (`POST .../materiality/approve`).

Validate - [ ] Auditor **cannot** approve (RBAC default-deny holds for the approval action). - [ ] Saved assessment shows server-**computed** overall / performance / trivial amounts (computed from benchmark amount x percentages, not free-typed). - [ ] After approval the record is **locked**: editing (`PUT`) returns *"Materiality is approved and locked; reopen it before editing"* (409), and the screen shows it read-only with approver + timestamp. - [ ] Re-approving an already-approved assessment is rejected (409). - [ ] (Optional) Partner/Admin **Reopen** (`POST .../materiality/reopen`) returns it to draft and is recorded in the audit trail.

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3. Risks - document, then load the ISA program and link

Screens: Risks (`/engagements/:id/risks`) and **Audit program** (`/engagements/:id/program`). Both editable by Admin/Partner/Manager (create/patch); all roles can read.

Do 1. On **Risks**, create at least: - One **entity-level** risk (no FS area, no assertions) - e.g. "Management override of controls", inherent **High**, control **Medium**, mark **significant**. - One **assertion-level** risk tied to an FS area - e.g. Revenue, assertions **Occurrence/Accuracy**, inherent **High**, control **High**. (`POST .../risks`) 2. On **Audit program**, click **"Load standard ISA program"** - this seeds the firm template library (idempotent) and bulk-adds procedures to the engagement in one action (`POST .../procedures/from-templates`). No manual seed step needed. 3. Link procedures to risks: on a risk-response procedure (e.g. a revenue procedure), set its linked risks (`PUT .../procedures/{id}/risks`) to the assertion-level revenue risk. 4. Set a procedure's status and assignee/due date (`PATCH .../procedures/{id}` and `.../status`).

Validate - [ ] Entity-level risk saved with **no** FS area/assertions; assertion-level risk saved **with** FS area + assertions (the model enforces this split). - [ ] **Combined rating is derived, not entered**, per the two-factor matrix: - inherent High x control Medium -> combined **High** (the entity risk above). - inherent High x control High -> combined **High** (the revenue risk). - (Spot-check one LowxLow -> **Low** if you add a third risk.) - [ ] "Load standard ISA program" adds the procedure set; clicking it **again does not duplicate** (idempotent seed). - [ ] At least one procedure shows its **linked risk(s)**, and at least one risk shows a planned response/linked procedure - i.e. risks<->procedures are connected end-to-end. - [ ] Procedure status/assignee/due-date update persists and the assigned procedure appears under **My procedures** for that assignee.

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4. Lead schedules - totals reconcile to the TB

Screen: Lead schedules (`/engagements/:id/lead-schedules`). `GET .../lead-schedules` (list) and `.../lead-schedules/{fs_area}` (detail).

Do 1. Open Lead schedules - one schedule auto-generated per FS area from the imported TB. 2. Open the **Revenue** and **Cash and bank** detail schedules.

Validate (against the sample TB) - [ ] Each FS-area schedule's **balance = sum of that area's TB lines**, e.g.: - Revenue -> **-900,000** (PY -800,000, movement -100,000) - Cost of sales -> **540,000** (PY 480,000) - Cash and bank -> **100,000** (PY 90,000) - Trade receivables -> **150,000**; Inventory -> **200,000**; PPE -> **500,000** - Trade payables -> **-120,000**; Borrowings -> **-300,000**; Equity -> **-430,000** - [ ] **Sum of all lead-schedule balances = 0** (mirrors the balanced TB) - this is the headline "correct totals" reconciliation check. - [ ] **Movement = current** – **prior** on each schedule; matches by-hand arithmetic. - [ ] Before any TB is imported on a fresh engagement, the screen reports *"No trial balance imported for this engagement"* (404) rather than an empty success - confirms empty ≠ error.

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5. Cross-cutting checks (audit trail + RBAC)

- [ ] Every write above (TB import, FS-area regroup, materiality save/approve/reopen, risk create/update, procedure add/link/status) appears in the **Audit trail** screen (Admin/Partner) with who / what / when / before-after.
- [ ] The Auditor RBAC block in step 2 held (approval denied).

Sign-off

- [ ] All boxes above ticked -> **Phase 2 acceptance PASSED**; update `checklist.md` (Phase 2 status note) and proceed to Phase 3.
- [ ] If any box fails -> log a defect, fix on a feature branch + draft PR (CI green to merge), re-run the affected step. **Do not start Phase 3 until this passes.**

Field	Value

Engagement used	
Run by (Partner/Admin)	
Run by (Auditor/Manager)	
Date	
Result	PASS / FAIL
Notes / defects	